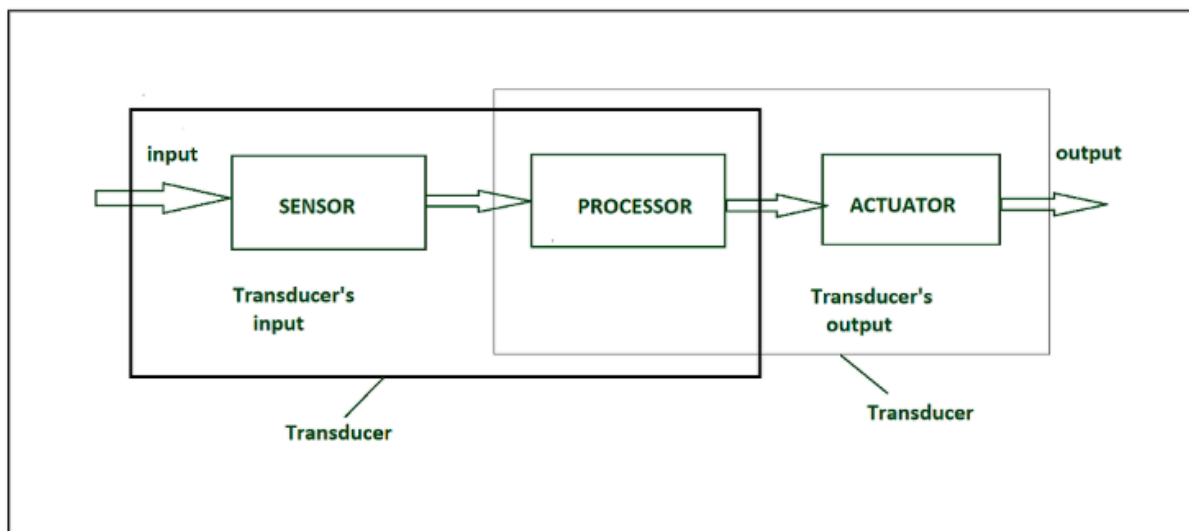


Sensor

A Sensor is a characteristic of any device or material to detect the presence of a particular physical quantity. The output of the sensor is a signal, which is converted to human readable form. It performs some function of input by sensing or feeling the physical changes in the characteristics of a system in response to stimuli.

What are Sensors?

Sensors are the devices that can detect and response to changes in the environment. These changes can be in form of light, temperature, motion, moisture or any other physical property. The sensor converts these physical changes into signal that can be measured. Sensors play an important role in IoT which will make an ecosystem for collecting, analyzing, and processing data about a specific environment so that it can be monitored, managed, and controlled more easily and efficiently. Sensors bridge the gap between the physical world and the logical world.



Transducer: It converts the signal from one physical form to another physical form. It is also called energy converter. For example, microphone converts sound to electrical signal. It is based on the principle of conservation of energy.

Classification of Sensors

The Sensor can be classified as

1. Based on Power Requirement

- **Active Sensors:** These sensors can *generate their own electrical signal* in response to physical changes and do not require external excitation power

Mechanism: Emits energy -> Interacts with target -> Detects backscatter/reflection.

Examples: Synthetic Aperture Radar (SAR), LiDAR, Laser fluorosensors.

LiDAR: <https://youtu.be/dOvMDbQKpwQ>

- **Passive Sensors:** These sensors *require an external power source (excitation)* and modify that signal according to the physical quantity being measured.

Passive sensors detect natural radiation, either reflected from the sun or emitted by the Earth (thermal infrared)

Mechanism: Detects ambient/naturally occurring energy.

Examples: Radiometers, spectrometers, photographic cameras (without flash).

2. Based on Means of Detection

The Sensors can be according to the detection method they use such as electrical, biological, chemical, or radioactive detection.

3. Based on Measurement



Temperature Sensors

- Measure heat or cold
- Examples: **Thermocouple, RTD, Thermistor**
- RTD most commonly stands for Resistance Temperature Detector, a type of sensor that measures temperature by detecting changes in electrical resistance, using materials like platinum for high accuracy. Resistance increases with increase in temperature.

●



Light Sensors

- Detect light intensity
- Examples: **LDR, Photodiode, Phototransistor**
- A Light Dependent Resistor (LDR), or photoresistor, is a passive electronic component whose resistance decreases as light intensity increases, based on the principle of photoconductivity. Widely used for automatic lighting (streetlights), cameras, and alarms, they feature high resistance in darkness (megaohms) and low resistance in bright light (hundreds of ohms)



Sound Sensors

- Detect sound waves
- Examples: **Microphone**



Proximity Sensors

- Detect presence or distance without contact
- Examples: **Inductive, Capacitive, Ultrasonic, Infrared**



Pressure Sensors

- Measure gas or liquid pressure
- Examples: **Strain gauge, Piezoelectric**



Position Sensors

- Detect object position or movement

- Examples: **Potentiometer, Encoder** (Encoder sensors are a type of mechanical motion sensor that create a digital signal from a motion.)

⚡ **Current & Voltage Sensors**

- Measure electrical parameters
- Examples: **Hall-effect sensor, Voltage divider**

🌫️ **Gas Sensors**

- Detect presence of gases
- Examples: **MQ-series gas sensors**

💧 **Humidity Sensors**

- Measure moisture in air
- Examples: **Capacitive humidity sensor**

🚶 **Motion Sensors**

- Detect movement
- Examples: **PIR (Passive Infrared) sensor**

🧲 **Magnetic Sensors**

- Detect magnetic fields
- Examples: **Hall-effect sensor, Magnetometer**

4. Based on the Conversion Phenomenon

This classification is based on the input and output conversion

- Photoelectric: It Changes light to electrical signals.
- Thermoelectric: It Changes temperature difference to electrical voltage.
- Electrochemical: It Changes chemical reactions to electrical signals.
- Electromagnetic: It Changes magnetic fields to electrical signals.
- Thermoptic: It Changes temperature changes to electrical signals.

5. Based on Output Type

- **Analog Sensors:** It produce an output signal which is usually in the form of voltage, current, or resistance, proportional to the measured quantity.
- **Digital Sensors:** It provide discrete or digital data as output.

❖ **Application of Types Of Sensors**

We live in the world of sensors. There are different types of sensors in our homes, offices, cars etc. by working to make our lives easier by turning on the lights by detecting our presence, adjusting the room temperature, detect smoke or fire, make us delicious coffee, and automatic door closing and so on. here we will discuss applications of sensors one by one in detail:

Given below are the Application of Types Of Sensors

- **Automotive Industry:** They are used in the Automotive industry for monitoring engine temperature, speed and other parameters.

- **Smart Homes:** They are used in the Smart Homes for detecting movements, Control HVAC (Heating, Ventilation, and Air Conditioning) and other measurements.
- **Robotics:** They are used in the [Robotics](#) for object recognition, Tracking the position and measuring force.
- **Transportation:** Sensors such as [GPS](#), Load, and Speed sensors are used in transportation infrastructure.

❖ Key Parameters of Sensors

Sensitivity

Sensitivity is the amount by which a sensor's output changes as the quantity being measured changes. *A sensor that is very sensitive can pick up on small changes, giving more precise and accurate information.*

For example, a very sensitive temperature sensor could identify a change in temperature as small as a fraction of a degree, making it perfect for precise applications like climate control systems in labs.

✅ General Formula for Sensitivity

$$\text{Sensitivity} = \frac{\text{Change in Output}}{\text{Change in Input}}$$

or mathematically,

$$S = \frac{d(\text{Output})}{d(\text{Input})}$$

◆ Examples

1. Temperature Sensor

$$S = \frac{dV}{dT} \quad (V/^{\circ}C)$$

2. Displacement Sensor

$$S = \frac{dC}{dx} \quad (F/m)$$

3. Pressure Sensor

$$S = \frac{dV}{dP} \quad (V/Pa)$$

4. Strain Gauge

$$S = \frac{dR}{d\varepsilon}$$

Range

The **minimum and maximum values** of a physical quantity that a sensor can measure are called its range.

For example, *the pressure gauge in a weather station should have a range that includes the lowest and highest pressures it could experience.*

By knowing a sensor's range, you can make sure it works within its limits, avoiding wrong readings or possible damage.

Accuracy

Accuracy is how close the reading from the sensor is to the true value of the physical quantity.

For example, *a blood glucose meter needs to be very accurate to give correct readings so that the right treatment choices can be made.*

Resolution

Resolution is the smallest change that a sensor can pick up in a physical quantity.

A sensor with a high precision can tell the difference between readings that are very small.

For example, a high-resolution weight sensor could identify even a one-gram difference, which is important for applications that need precise measurements, like laboratory scales or jewelry weighing machines.

❖ **Categories of Sensors:**

Sensors are devices that detect physical changes and convert them into measurable signals.

Here are five major categories:

1. Resistive sensors
2. capacitive sensors
3. inductive sensors
4. optical sensors
5. piezoelectric sensors

1. Resistive Sensors

How They Work: Electrical resistance changes when exposed to physical stimuli.

Common Types:

- **Thermistors** - Temperature sensing
- **Strain Gauges** - Force and weight measurement (When a material is stretched or compressed, its electrical resistance changes.)
- **Potentiometers** - Position detection
- **Photoresistors (LDR)** - Light intensity measurement
- **Force Sensing Resistors (FSR)** - Pressure detection

Applications:

- Temperature controllers and thermostats
- Load cells and weighing scales
- Volume controls and joysticks
- Automatic lighting systems

- Touch-sensitive devices

Examples

1. Strain gauge :

A **strain gauge** is a versatile and widely used sensing device that **measures strain or deformation** in various materials. It is a fundamental component in the field of engineering and plays a crucial role in measuring and monitoring mechanical forces and stresses. Strain gauges are utilized in a wide range of applications, including *structural analysis, load testing, material testing, and stress analysis*.

The basic principle behind a strain gauge is its ability to convert mechanical strain into an electrical signal. The gauge consists of a thin metallic wire that is attached to the surface of the material under observation. When the material is subjected to strain, the gauge undergoes deformation, which alters the electrical resistance of the metallic element. This change in resistance is directly proportional to the applied strain, allowing for accurate measurements.

What is Strain Gauge?

A **strain gauge** is a device used to measure the strain or deformation of an object under external forces. It is widely employed in the field of **solid mechanics to quantify the amount of stress experienced by a material**. The gauge consists of a wire or a foil patterned in a zigzag or grid-like configuration, which undergoes changes in its electrical resistance when subjected to mechanical strain. Strain gauges find extensive applications in various fields, including structural engineering, **aerospace, automotive, and materials testing**. They are used to monitor and analyze the behavior of structures, determine material properties, assess structural integrity, and optimize designs.

Strain Gauge Working Principle

Strain gauge working principle is based on the concept of electrical resistance variation due to mechanical strain. The strain gauge consists of a fine wire or foil, typically made of a highly conductive material like copper or constantan, which is attached to a flexible backing material. When a mechanical strain is applied to the strain gauge, the wire or foil undergoes deformation, resulting in a change in its length and cross-sectional area. This alteration in dimensions causes a corresponding change in the electrical resistance of the wire or foil.

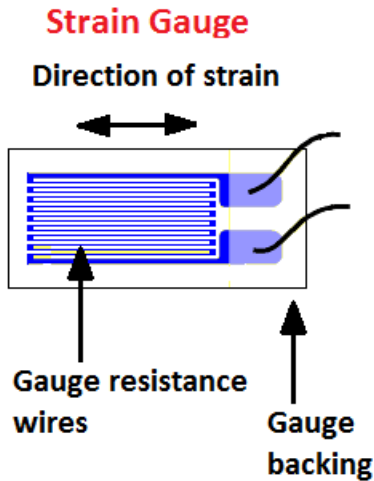


Figure #1

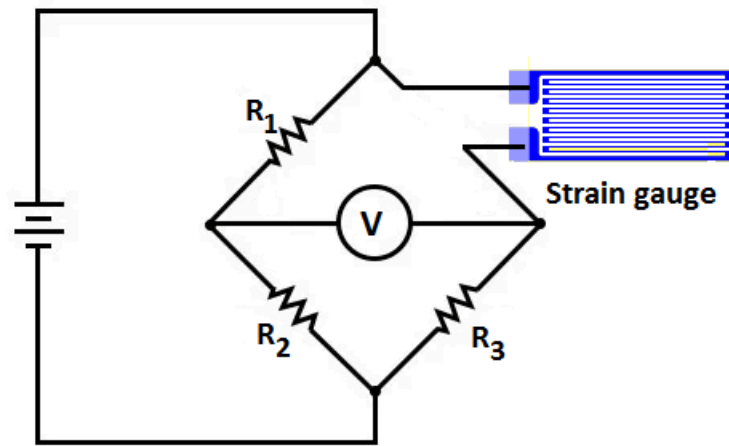


Figure #2

FIGURE: Schematic representation of strain gauge

The change in resistance can be measured using a Wheatstone bridge circuit, which is commonly employed in strain gauge applications. The Wheatstone bridge consists of four resistive arms, with the strain gauge forming one of the arms. The other three arms typically consist of fixed resistors. When the strain gauge experiences a strain, the resistance of the gauge changes, leading to an imbalance in the Wheatstone bridge circuit. This imbalance generates a small electrical output signal, which is proportional to the applied strain.

By measuring the output signal, the magnitude of the strain can be determined. This allows for the analysis of various mechanical properties such as stress, load, and deformation in structural components. Strain gauges find extensive applications in areas like civil engineering, aerospace, automotive, and material testing. We will learn more about its applications in civil engineering in following sections.

Applications of Strain Gauge

The **applications of strain gauges** in the field of civil engineering are invaluable. Strain gauges are extensively used *to measure and monitor the deformations and stresses experienced by various structural components and materials*. These devices play a crucial role in ensuring the safety, durability, and efficiency of civil engineering projects.

One significant application of strain gauges is in the field of **structural health monitoring**. By attaching strain gauges to critical sections of structures such as bridges, dams, and

buildings, engineers can accurately measure the strains and stresses these structures endure under normal and extreme conditions.

Strain gauges are also used in experimental studies and material testing. In SOM, researchers often apply strain gauges to test specimens and structural models *to study the behavior of materials under different loads and environmental conditions*. By measuring the strains in real-time, engineers gain insights into the material properties, stress distributions, and failure mechanisms, aiding in the development of safer and more efficient designs.

Strain gauges also have application in **geotechnical engineering**. They are employed *to monitor the deformations in soil, rock, and underground structures such as tunnels and retaining walls*. By measuring the strains caused by ground movements, engineers can assess the stability of slopes, predict potential failures, and implement appropriate measures to prevent disasters.

Video link: [📺 Working of Strain Gauges and their Applications | Skill-Lync](#)

Advantages of Strain Gauge

1. **High Sensitivity:** Strain gauges have excellent sensitivity, allowing for accurate measurement of even minute deformations. This feature is crucial in civil engineering, as it helps detect structural movements or stress concentrations that may affect the integrity of a building or infrastructure.
2. **Versatility:** Strain gauges can be used on various materials such as concrete, steel, or composites, making them versatile for different civil engineering applications. They can be attached to surfaces or embedded within structures, enabling engineers to monitor strain at specific locations of interest.
3. **Cost-Effective:** Strain gauges offer a cost-effective solution for strain measurement compared to other techniques such as extensometers. They are relatively inexpensive and reusable, allowing for multiple measurements at different locations within a structure.
4. **Real-Time Monitoring:** Strain gauges provide real-time data, allowing engineers to monitor structural behavior continuously. This feature is particularly useful during load testing, construction, or seismic events, as it enables prompt identification of any excessive strains or deformations.

Disadvantages of Strain Gauge

1. **Installation Complexity:** The installation of strain gauges can be a time-consuming and intricate process, requiring expertise and precision. Specialized knowledge is

necessary to ensure proper attachment, wiring, and calibration of the strain gauges, which can increase the overall project complexity.

2. **Susceptible to Environmental Factors:** Strain gauges are sensitive to environmental conditions such as temperature, humidity, and vibration. These factors can introduce additional strains or interfere with the accuracy of measurements. Proper environmental controls and shielding techniques are necessary to mitigate these influences.
3. **Limited Measurement Range:** Each strain gauge has a specific measurement range, beyond which it may not provide accurate readings. This limitation restricts their application in cases where high strains are expected, such as during dynamic load testing or extreme events like earthquakes.
4. **Vulnerability to Damage:** Strain gauges are delicate and can be easily damaged during construction activities or due to accidental impacts. Protecting them from mechanical damage is crucial to ensure reliable and consistent measurements.

2. Potentiometer:

In the context of electrochemistry and sensors, a potentiometer used with ion-selective membranes does not change its mechanical resistance like a volume knob.

Instead, contact with ions causes a change in the potentiometric potential (voltage) or the membrane impedance (resistance), which is used to measure ionic concentration.

When a solid-contact ion-selective electrode (ISE) makes contact with ions, the following changes occur:

1. **Change in Membrane Resistance (Ohmic Resistance) Ion Binding:** Ion-selective membranes contain ionophores that selectively bind with target ions, increasing the ion concentration within the membrane.

Resistance Decrease: As ions enter the membrane, its ionic conductivity increases, which lowers the electrical bulk resistance of the membrane.
Thin Membrane Behavior: In solid-contact sensors with thin membranes (e.g., 5-10 micrometer), this resistance change can be significant and rapid, allowing for faster response times.

2. **Change in Potential (Voltage Generation) Nernstian Response:** The contact with ions creates a potential difference (voltage) across the membrane, which is directly proportional to the logarithm of the ionic activity (concentration) in the solution.
Voltage Shift: As ion concentration changes, the membrane potential shifts according to the Nernst equation, allowing the potentiometer to detect changes in ion concentration.

2. Capacitive Sensors

How They Work: Capacitance changes when the distance between plates or dielectric material changes.

Common Types:

- **Capacitive Proximity Sensors** - Detect nearby objects
- **Capacitive Touch Sensors** - Touch detection
- **Capacitive Pressure Sensors** - Pressure measurement
- **Capacitive Humidity Sensors** - Moisture detection

Key Feature: Can detect both metal and non-metal objects (plastic, liquid, wood)

Applications:

- Smartphone and tablet touchscreens
- Proximity detection in automation
- Liquid level sensing in tanks

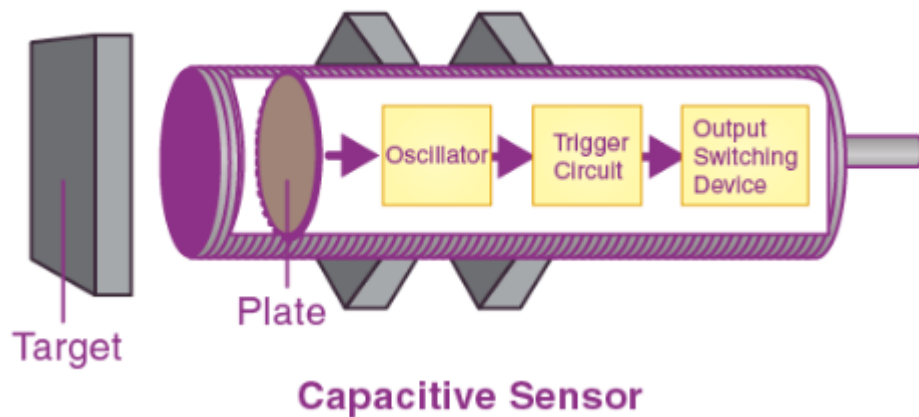
What is a Capacitive Sensor?

The non-contact device that can detect or sense the presence or absence of any object virtually irrespective of its material is known as a capacitive proximity sensor.

They use the alteration of capacitance based on the change in the electrical field and the electrical property of capacitance and everywhere around the active face of the sensor.

How a Capacitive Sensor Works

1. A capacitive sensor has two conductive plates separated by a small gap or insulating material.
 2. The space between plates may be air, liquid, plastic, or any dielectric material.
 3. When something comes near or changes between the plates, the capacitance changes.
 4. The sensor detects this change in capacitance and converts it into an electrical signal.
- *Capacitive sensors can sense anything. The object should have a dielectric different from that of air, or it must be conductive; the only conditions for the detection by capacitive sensing.*
 - This technique is used by many types of sensors. The main applications consist of the sensors used to detect position or displacement, proximity, fluid level, humidity and acceleration. Capacitive sensors can also replace mechanical buttons. Capacitive sensing touch screens as input devices are used by some modern technical devices such as digital audio players, mobile phones & tablet computers.
 - **Working Principle of a Capacitive Sensor**
 - Along with a sensing surface formed by two metal electrodes, the capacitive proximity sensor involves a high-frequency oscillator. When an object comes near the sensing surface, it changes the capacitance of the oscillator by entering into the electrostatic field of the electrodes. Due to this, **the oscillator circuit begins to oscillate** and varies the output state of the sensor when it reaches a certain amplitude. The oscillator's amplitude decreases as the object moves away from the sensor, switching the sensor back to its initial state.



-

Figure: Schematic diagram of the capacitive sensor

- If the value of the dielectric constant of a target is larger, then it becomes easier for the capacitive proximity sensor to detect it. Regardless of the type of material being sensed, they naturally have a short sensing range of about 1 inch.
- With an increase in the sensing surface size of the sensor, the surface area of the target and the dielectric constant of the target, the sensing distance increases while dealing with non-conductive targets.
- Humidity monitoring

Video link: [YouTube Proximity Sensor working. Inductive proximity sensor, capacitive proxim...](#)

3. Inductive Sensors

How They Work: Generate electromagnetic field; detect metallic objects that disturb this field.

Metal objects are detected in an array of industrial applications, using inductive proximity sensors. They work by setting up a magnetic field and then noticing any changes caused by the presence of the object. That is to say, when a metallic object enters the sensing area, it interferes with the electromagnetic field emitted from the front face of the sensor, resulting in a change in the total inductance. The change is recognized by this sensor, which then initiates the appropriate response.

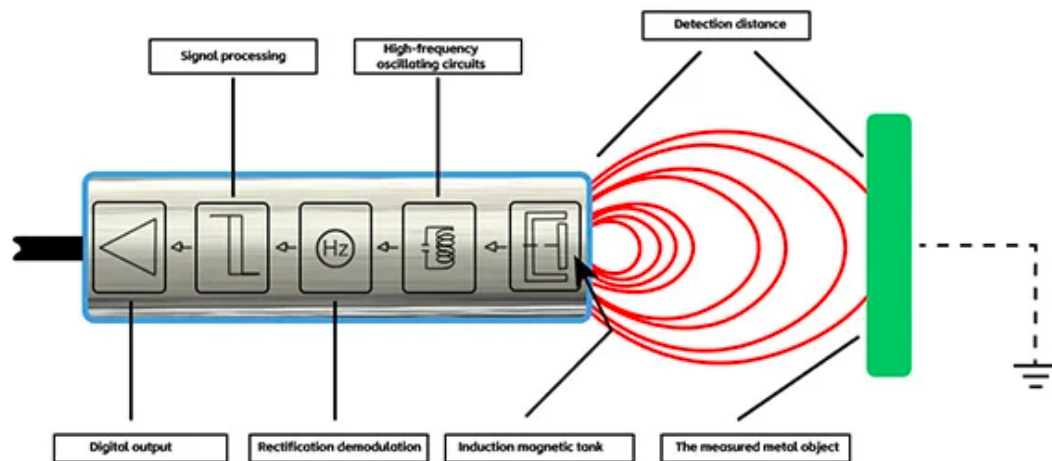


Figure: Schematic diagram of the inductive sensor

Working Principle

1. **Electromagnetic Field Generation:** An internal coil and oscillator circuit create a high-frequency, oscillating magnetic field that projects from the sensor's face.
2. **Eddy Current Induction:** When a metallic object (like iron, aluminum, copper) moves into this field, the alternating magnetic energy induces small, circular electric currents, called eddy currents, within the target.
3. **Energy Absorption:** These eddy currents absorb energy from the sensor's magnetic field, effectively dampening or reducing the oscillation's amplitude and/or frequency.
4. **Detection & Switching:** A detector circuit monitors these changes in oscillation. When the change crosses a set threshold (indicating a metal object is present), the circuit triggers an output switch.
5. **Output Signal:** The sensor then sends a signal (e.g., ON or OFF, depending on its configuration) to control other devices, like PLCs, relays, or lights.

<https://www.balluff.com/en-us/blog/the-basic-operating-principle-of-an-inductive-proximity-sensor>

Common Types:

- Inductive Proximity Sensors - Presence detection
- LVDT (Linear Variable Differential Transformer) - Precise displacement
- Eddy Current Sensors - Non-contact measurement

Key Feature: Detect **metals only**, highly resistant to dirt and moisture

Applications:

- Metal detection in security
 - Position sensing in machines
 - Object counting on conveyor belts
 - Speed detection in vehicles
 - Traffic light vehicle detection
-

4. Optical Sensors

How They Work: Use light (visible, infrared, or laser) to detect changes; convert light into electrical signals.

Common Types:

- **Photodiodes/Phototransistors** - Light intensity measurement
- **Through-Beam Sensors** - Object detection via beam interruption
- **Fiber Optic Sensors** - Flexible light transmission
- **Laser Sensors** - High-precision distance measurement
- **Image Sensors (CCD/CMOS)** - Image capture
- **Color Sensors** - Color detection

Key Feature: Non-contact, high precision, versatile

Applications:

- Barcode scanners and QR readers
- Automatic doors and motion sensors
- Smartphone ambient light sensors
- LIDAR in autonomous vehicles
- Machine vision and quality inspection
- Medical pulse oximeters

What is a photodiode?

Photodiodes are semiconductor devices that convert light into electrical current or voltage. They are a type of photodetector specifically designed for the detection of light. They are commonly used in various applications such as solar cells, optical communication systems, and light detection sensors.

Photodiodes are made from materials like silicon, germanium, or other semiconductor compounds that exhibit the properties of both conductors and insulators. They operate by exploiting the photoelectric effect, in which photons of sufficient energy that are incident on a semiconductor material can promote electrons from the valence band to the conduction band, resulting in the creation of electron-hole pairs.

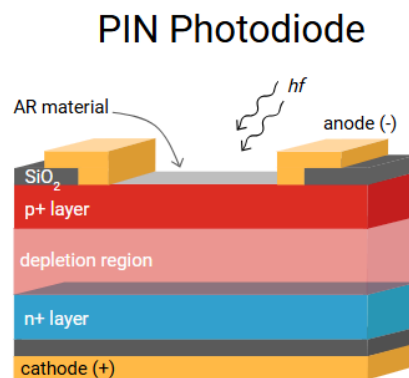


Figure: Schematic representation of PIN photodiode

5. Piezoelectric Sensors

How They Work: Generate electric charge when mechanical stress is applied (and vice versa).

Piezoelectricity, also called the piezoelectric effect, is the ability of certain materials to generate an AC (alternating current) voltage when subjected to mechanical stress or vibration, or to vibrate when subjected to an AC voltage, or both.

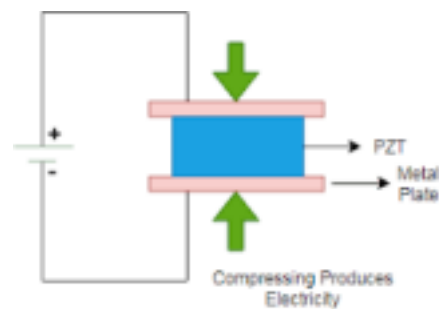
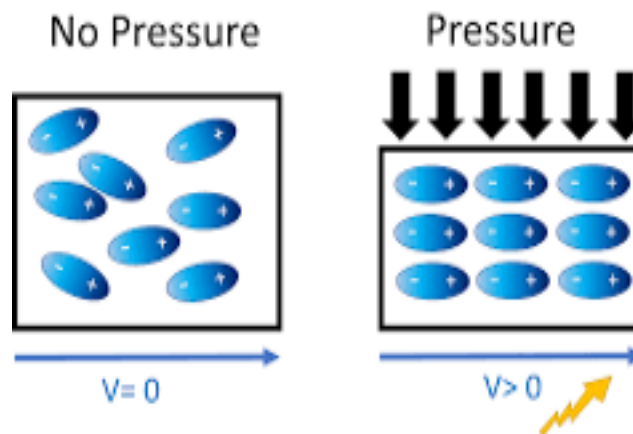
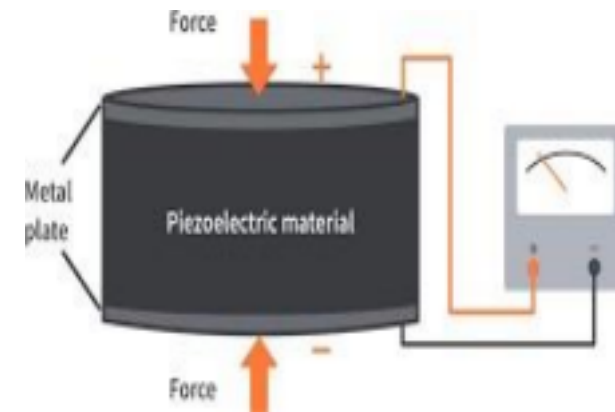
Piezoelectric sensors operate on the [piezoelectric effect](#), generating an electrical charge in response to applied mechanical stress (pressure, force, or acceleration) on materials like quartz or specialized ceramics.

A Piezoelectric Sensor does not require an external voltage or current source; it may generate an output signal based on the strain exerted.

Piezoelectric sensors are devices that transform mechanical energy (such as pressure, force, or vibration) into electrical energy. It is based on the piezoelectric effect, a phenomenon in which some materials generate an electric charge when deformed.

Principle of piezo-electricity

Piezoelectricity is linear interaction between mechanical and electrical systems in non-centric crystals or similar structures



Applications:

- Vibration monitoring in machines
- Airbag sensors in cars
- Ultrasound medical imaging
- Electric guitar pickups

- Engine knock sensors
- Touch-sensitive smartphone screens
- Quartz watches (oscillator)

Common Piezoelectric Materials:

- Quartz crystals
- PZT ceramics (Lead Zirconate Titanate)
- PVDF polymers

Common Types:

- **Piezoelectric Accelerometers** - Vibration and acceleration
- **Piezoelectric Pressure Sensors** - Dynamic pressure
- **Piezoelectric Force Sensors** - Force measurement
- **Acoustic Sensors** - Sound and ultrasound detection

Key Feature: Self-generating (no external power needed), excellent for dynamic measurements

Smart Materials

Smart Materials: Smart materials are materials that are manipulated to respond in a controllable and reversible way, modifying some of their properties as a result of external stimuli such as certain mechanical stress or a certain temperature, among others.

Why smart materials are used?

Smart materials and systems are able to sense and respond to the environment around them. • They have the potential to improve existing technology and add new functionality to products. • They have applications in a wide variety of areas and could have an important role in waste reduction.

1. Photodiode Materials:

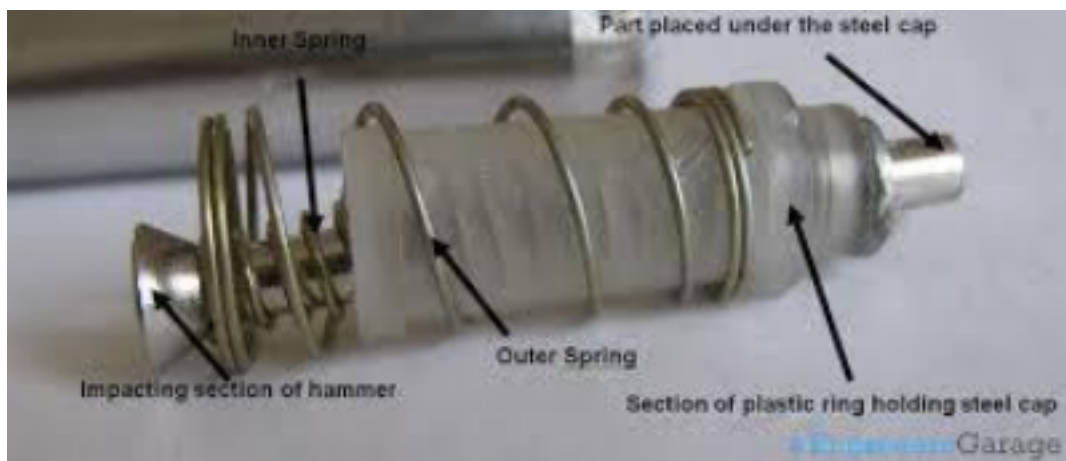
Photodiode is a pn junction of a semiconductor material that produces electron hole pairs in response to light. Photodiodes primarily use semiconductor materials like Silicon (Si), Germanium (Ge), and Indium Gallium Arsenide (InGaAs) to convert light into electrical energy via the photovoltaic effect. Silicon is standard for visible/NIR light (190–1100 nm), while Ge and InGaAs cover infrared (up to 2600 nm). Other materials include Lead Sulfide, Mercury Cadmium Telluride, and newer options like graphene.

Key Photodiode Materials and Spectral Ranges:

- Silicon (Si) (190–1100 nm): Most common material for visible and near-infrared light; low noise.
- Germanium (Ge) (400–1700 nm): Used for infrared detection, but has higher noise than silicon.
- Indium Gallium Arsenide (InGaAs) (800–2600 nm): Preferred for high-speed fiber-optic communication and infrared sensing.
- Mercury Cadmium Telluride (HgCdTe) (400–14000 nm): Used for long-wavelength infrared (LWIR) applications, often requiring cooling.
- Silicon Carbide (SiC): Suitable for UV and high-temperature environments.
- Material Selection Criteria: Material choice is determined by the required wavelength, sensitivity, and response time. For example, Si is used for visible light, whereas InGaAs is used for infrared. The material must have a bandgap that matches the photon energy of the target light. Newer materials such as graphene and MoS_2 are also being utilized.

2. Piezoelectric Materials:

The common application of piezoelectric material is a gas igniter. (It consists of a small, spring-loaded hammer which, when a button is pressed, hits a PZT crystal of PZT. This sudden forceful deformation produces a high voltage and subsequent electrical discharge, which ignites the gas.



Forms of Piezoelectric Material

- ☐ Quartz (SiO_2)
- ☐ Lithium Niobate (LiNbO_3)

- Lead Zirconate Titanate PZT

Role of Piezoelectric material in Civil Infrastructure

- Determination of structural damage that is presented in structure in the RCC structure
- Determination of location of damage
- Quantification of severity of the damage
- Prediction of the remaining service life of the structure

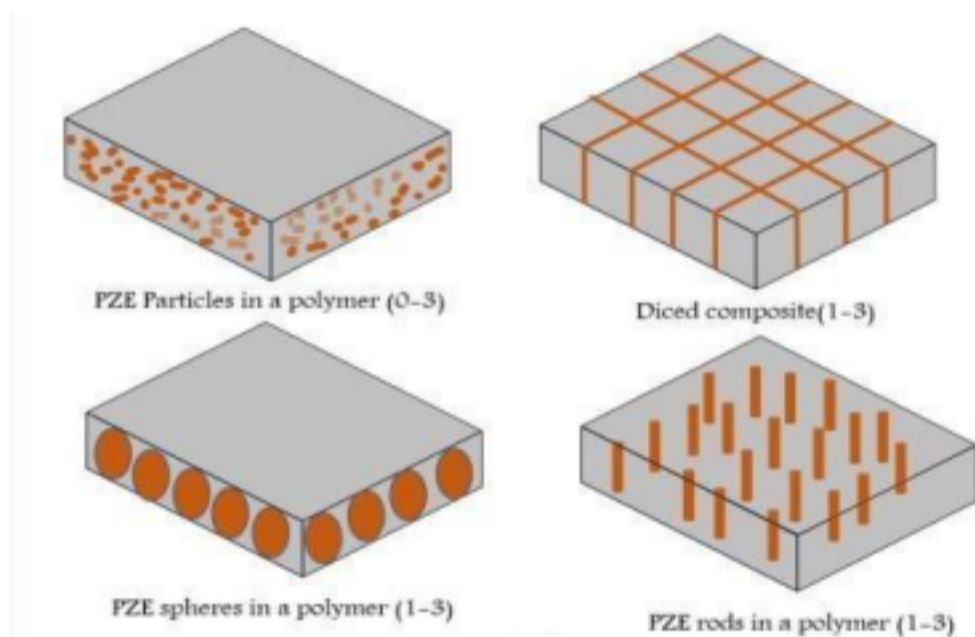
Piezo-electric materials (Polymers and Ceramics)

Piezoelectric materials can be classified in three categories:

- Piezoelectric polymers
- Piezoelectric ceramics
- Piezoelectric composites

A. Piezoelectric polymers

Piezoelectric polymers are polymers that can generate electric charges on the surface under pressure/strain thus converting mechanical energy into electrical energy. The most commonly explored piezoelectric polymers are poly (vinylidene fluoride) (PVDF) or its copolymer poly (vinylidene fluoride-co-trifluoroethylene) (PVDF-TrFE).

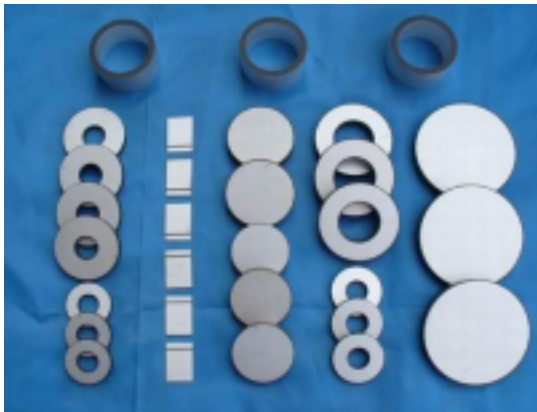


B. Piezoelectric Ceramics

A piezoelectric ceramic is a smart material that converts a mechanical effect (such as pressure, movement, or vibration) into an electrical signal and vice versa. These are commonly used in mechatronic devices and smart systems. They can be used to sense changes in pressure and strain in the environment and can generate electrical responses.

The most commonly produced piezoelectric ceramics are lead zirconate titanate (PZT), barium titanate, and lead titanate.

To prepare a piezoelectric ceramic, fine PZT (lead zirconate titanate) powders of the component metal oxides are mixed in specific proportions, then heated to form a uniform powder. The piezo powder is mixed with an organic binder (water, oil, sodium silicate, bentonite and corn starch) and is formed into structural elements having the desired shape (discs, rods, plates, etc.).



3. Piezoelectric composites

Piezoelectric composites are materials that combine the piezoelectric properties of certain ceramics or polymers with other materials to create enhanced, multifunctional materials. Piezoelectricity refers to the ability of certain materials to generate an electric charge in response to mechanical stress. This effect is reversible, meaning that applying an electric field can also induce mechanical deformation.

B. Installation of Piezoelectric material in concrete structure

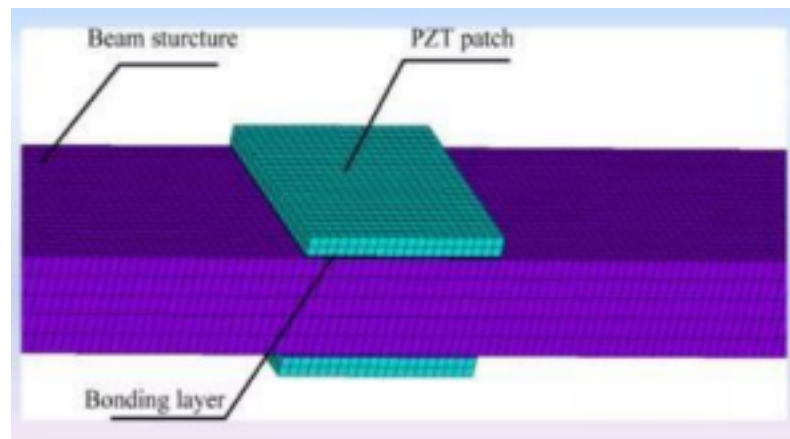
For the assessment of active response of the structure the piezoelectric material can be

installed in two ways

1. Surface Bonded

Bonded to the surface of concrete through a high strength epoxy adhesive. ·

Drawback material rapidly deteriorates with respect to atmospheric weather conditions.



Embedded piezoelectric material

- ☐ Install at the time of construction structure
- ☐ Safe against the aggressive environmental condition
- ☐ More effective as compare to the surface bonded material.



3. Strain Gauge Material

A strain gauge is a sensor that converts mechanical displacement into electrical signals that can be measured.

A strain gauge is a thin, waferlike device that can be attached to a variety of materials to measure applied strain.

Sensing element/material used in strain gauges:

Among the commonly used materials for strain gauge sensing elements, **Constantan** is the most widely used due to its superior characteristics.

Why Constantan is Used

- **Composition:**
Constantan is an alloy composed primarily of copper (approximately 55%) and nickel (approximately 45%). This unique composition provides properties that make it ideal for strain gauge applications.
- **Stability of Electrical Resistance:**
Constantan exhibits excellent stability in electrical resistance over a wide range of temperatures, ensuring reliable strain measurements even under varying environmental conditions.
- **Low Thermal Coefficient of Resistance:**
Constantan has a very low temperature coefficient of resistance, meaning its resistance changes minimally with temperature. This significantly reduces temperature-induced measurement errors.

- **High Sensitivity:**
The alloy has good sensitivity to strain, enabling accurate detection of small deformations in the material to which it is bonded.
 - **Durability:**
Constantan is mechanically robust and resistant to environmental effects, making it suitable for long-term and repeated use.
 - **Ease of Manufacturing:**
Constantan can be easily fabricated into fine wires or thin foils, which are commonly used as sensing elements in strain gauges.
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